

Evaluate

$$\begin{aligned}
 \textcircled{1} \int_0^1 \left(\frac{4}{t^2+1} \right) dt &= 4 \arctan(t) \Big|_0^1 \\
 &= 4(\arctan 1 - \cancel{\arctan 0}) \\
 &= 4\left(\frac{\pi}{4} - 0\right) = \pi \rightarrow
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{2} \int_{-1}^1 e^{u+1} du &= e^{u+1} \Big|_{-1}^1 \\
 &= e^2 - e^0 \\
 &= e^2 - 1
 \end{aligned}$$

Definite integral!

Effectiveness of FTC depends on having a good supply of antiderivatives

Indefinite integral

$$\int f(x) dx = F(x) + C \quad \text{if and only if} \\
 F'(x) = f(x)$$

$$\int_1^3 x^3 dx = \frac{1}{4} x^4 \Big|_1^3 = \frac{81}{4} - \frac{1}{4} = 20 \quad (\text{Number})$$

$$\int x^3 dx = \frac{1}{4} x^4 + C \quad (\text{Function})$$

Family of functions.

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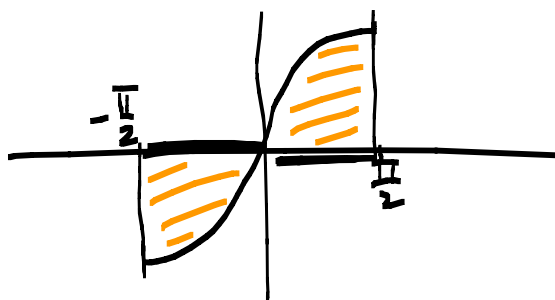
Three types questions:

$$1. \int \sin x \, dx = -\cos x + C$$

$$2. \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x \, dx = -\cos x \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\ = -\cos\left(\frac{\pi}{2}\right) - \left(-\cos\left(-\frac{\pi}{2}\right)\right) = 0$$

3. Find the area between the graphs of $y = \sin x$, the x -axis and $x = \frac{\pi}{2}$, $x = -\frac{\pi}{2}$

Sketch:



$$\left. \begin{aligned} \int_{-\frac{\pi}{2}}^0 \sin x \, dx &= -1 \\ \int_0^{\frac{\pi}{2}} \sin x \, dx &= 1 \end{aligned} \right\} \Rightarrow \text{Area is } 2$$

Evaluate the following:

$$\int x^5 \, dx = \frac{1}{6} x^6 + C$$

$$\int (1-2x)^5 \, dx = -\frac{1}{12} (1-2x)^6 + C \quad \left| \begin{array}{l} \frac{d}{dx} x^6 = 6x^5 \\ \frac{d}{dx} (1-2x)^6 \end{array} \right.$$

$$\int \sin^5 2x \cos 2x \, dx$$

$$= \frac{1}{12} \sin^6 2x + C$$

$$= 6(1-2x)^5 \cdot -2$$

$$= -12(1-2x)^5$$

$$\frac{d}{dx} \sin^6 2x$$

$$= 6 \sin^5 2x \cdot \cos 2x \cdot 2$$

$$1. \int [f(x)]^n f'(x) \, dx = \frac{[f(x)]^{n+1}}{n+1} + C$$

$$\frac{1}{2} \int \underbrace{(\sin 2x)^5 \cos 2x}_{\frac{d}{dx} \sin^6 2x} \, dx = \frac{1}{2} \frac{(\sin 2x)^6}{6} + C$$

$$= \frac{1}{12} \sin^6 2x + C$$

$$\int x \sqrt{1-x^2} \, dx = -\frac{1}{2} \int \underbrace{-2x}_{\frac{d}{dx} (1-x^2)} (1-x^2)^{1/2} \, dx$$

$$= -\frac{1}{2} \frac{(1-x^2)^{3/2}}{3/2} + C$$

$$= -\frac{1}{3} (1-x^2)^{3/2} + C$$

$$\frac{d}{dx} \left[-\frac{1}{3} (1-x^2)^{3/2} + C \right] = -\frac{1}{3} \cdot \frac{3}{2} (1-x^2)^{1/2} \cdot -2x$$

$$= x \sqrt{1-x^2}$$

$$\int e^{3t} dt = \frac{1}{3} e^{3t} + c$$

$$\frac{d}{dt} e^{3t} = 3 e^{3t}$$

$$2. \int e^{f(x)} f'(x) dx = e^{f(x)} + c$$

$$\int e^{\sin x} \cos x dx = e^{\sin x} + c$$

$$\frac{1}{6} \int e^{\tan 6x} \underbrace{6 \sec^2 6x} dx = \frac{1}{6} e^{\tan 6x} + c$$

$$\int e^{3x} \sec^2(e^{3x}) dx$$

$$3. \int \sec^2 f(x) \cdot f'(x) dx = \tan f(x) + c$$

$$\frac{1}{3} \int \underbrace{3 e^{3x} \sec^2(e^{3x})} dx = \frac{1}{3} \tan(e^{3x}) + c$$

Similar standard forms for all the trig functions!

$$\int \frac{1}{x} \sin(\ln x) dx = -\cos(\ln x) + c$$

$$-\int -\frac{1}{x^2} \cos\left(\frac{1}{x}\right) dx = -\sin\left(\frac{1}{x}\right) + c$$

$$-\frac{1}{2} \int \frac{(-2)}{1-2x} dx$$

$$\frac{d}{dx} \ln |1-2x| = \frac{-2}{1-2x}$$

$$= -\frac{1}{2} \ln |1-2x| + C$$

$$4. \int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\frac{1}{3} \int \frac{3 \sin 3x}{1 - \cos 3x} dx = \frac{1}{3} \ln |1 - \cos 3x| + C$$

$$\int \frac{1}{1 + \underline{x^2}} dx = \arctan x + C$$

$$\frac{d}{dx} \arctan \underline{bx} = \frac{b}{1 + (bx)^2}$$

$$5. \int \frac{f'(x)}{1 + [f(x)]^2} dx = \arctan [f(x)] + C$$

$$3 \int \frac{\cancel{3} 1}{1 + x^2} dx = 3 \arctan x + C$$

$$\int \frac{x}{1 + x^4} dx = \frac{1}{2} \int \frac{2x}{1 + (x^2)^2} = \frac{1}{2} \arctan x^2 + C$$

$$\begin{aligned}
 \int \frac{1}{(1-2x)^2} dx &= -\frac{1}{2} \int -2(1-2x)^{-2} dx \\
 &= -\frac{1}{2} \frac{(1-2x)^{-1}}{-1} + C \\
 &= \frac{1}{2} (1-2x)^{-1} + C \quad (\text{check!!})
 \end{aligned}$$

$$\begin{aligned}
 \int \frac{1}{\sqrt{1-2x}} dx &\stackrel{!}{=} -\frac{1}{2} \int -2(1-2x)^{-1/2} dx \\
 &= -\frac{1}{2} \frac{(1-2x)^{1/2}}{1/2} + C \\
 &= -\sqrt{1-2x} + C
 \end{aligned}$$

$$\begin{aligned}
 \int \frac{\sin 3x}{\sqrt{1-\cos 3x}} dx &= \frac{1}{3} \int 3 \sin 3x (1-\cos 3x)^{-1/2} dx \\
 &= \frac{1}{3} \cdot 2 (1-\cos 3x)^{1/2} + C
 \end{aligned}$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \int (1-x^2)^{-1/2} dx ??$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

$$6. \int \frac{f'(x)}{\sqrt{1-[f(x)]^2}} dx = \arcsin f(x) + C$$

$$\int \frac{1}{\sqrt{1-4x^2}} dx = \frac{1}{2} \int \frac{2}{\sqrt{1-(2x)^2}} dx$$

$$= \frac{1}{2} \arcsin(2x) + C$$

$$\int \frac{e^{3x}}{\sqrt{1-e^{6x}}} dx = \int e^{3x} (1-e^{6x})^{-\frac{1}{2}} dx$$

$$= \frac{1}{3} \left(\int \frac{3e^{3x} dx}{\sqrt{1-(e^{3x})^2}} \right)$$

$$= \frac{1}{3} \arcsin(e^{3x}) + \frac{1}{3} C$$

$$= \frac{1}{3} \arcsin(e^{3x}) + C_1$$

List of Standard Integrals

$$1. \int [f(x)]^n f'(x) dx = \frac{f(x)^{n+1}}{n+1} + C$$

$$2. \int e^{f(x)} \cdot f'(x) dx = e^{f(x)} + C$$

$$3. \int a^{f(x)} \cdot f'(x) dx = \frac{a^{f(x)}}{\ln a} + C, \quad a > 0$$

$$4. \int \sin f(x) \cdot f'(x) dx = -\cos f(x) + C$$

⋮

$$5. \int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

$$6. \int \frac{f'(x)}{1 + [f(x)]^2} dx = \tan^{-1}(f(x)) + C$$

$$7. \int \frac{f'(x)}{\sqrt{1 - [f(x)]^2}} dx = \sin^{-1}(f(x)) + C$$